रा.इ.सू.प्रौ.सं NIELIT
राष्ट्रीय इलेक्ट्रॉनिकी एवं सूचना प्रौद्योगिकी संस्थान, चेन्नई
National Institute of Electronics & Information Technology, Chennai

Ministry of Electronics & Information Technology
Government of India

नेशनल इंस्टीट्यूट ऑफ इलेक्ट्रॉनिक्स एंड इंफॉर्मेशन टेक्नोलॉजी, चेन्नई

National Institute of Electronics and Information Technology, Chennai

Autonomous Scientific Society of Ministry of Electronics & Information Technology (MeitY), Govt. of India

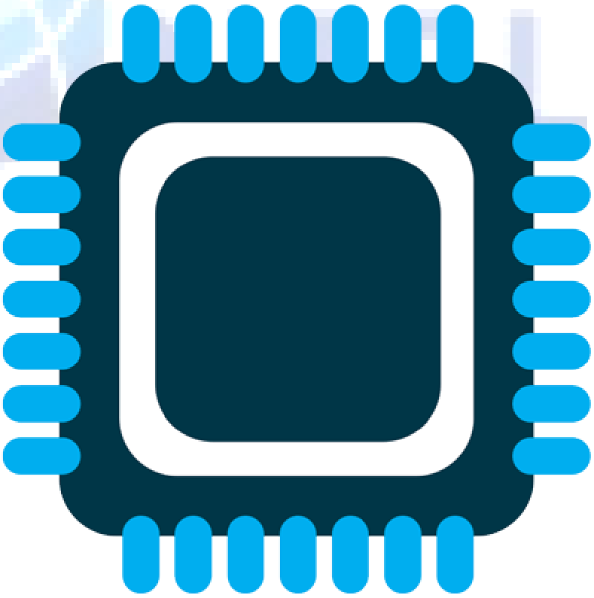
ISTE Complex, 25, Gandhi Mandapam Road, Chennai - 600025

Course Prospectus

Chip Design Associate (O-Level 'Chip Design')

Mode: Online

(NSQF-Level 4)



Index

Course Prospectus

Course Name: Chip Design Associate (O-Level 'Chip Design')

Course Code: VL151

NSQF Level: 4

Duration: 450 Hrs

Last Date of Registration: 22nd Sep 2025

Course Start Date: 24th Sep 2025

Fee Details:

Registration Fee- Rs. 1000/- (Adjusted with Total Fee)

Total Fee – Rs. 24,500/- (Fee is waived for SC/ST students)

Preamble:

The emergence of India as a global economy has opened up a huge demand for electronic products. National Policy on Electronics and Make in India initiative of the Government of India has resulted in the setting up of many industries in the Electronics sector and has led to a huge demand for trained manpower in the Embedded and VLSI (Very Large Scale Integration). With the development of semiconductor process technology, more and more transistors are integrated into a single chip. Consequently, Integrated Circuits (ICs) can now host more functionality on one chip, leading us to the System-On-Chip (SoC) era.

This course introduces the students to essential industry skills in the VLSI industry. The program covers Verilog HDL coding, Static Timing Analysis (STA), Field-Programmable Gate Array (FPGA) architecture, and FPGA emulation. Overall, the program prepares participants for diverse roles in Physical design and emulation projects, enhancing their employability and fostering industry innovation.

Objective of the Course:

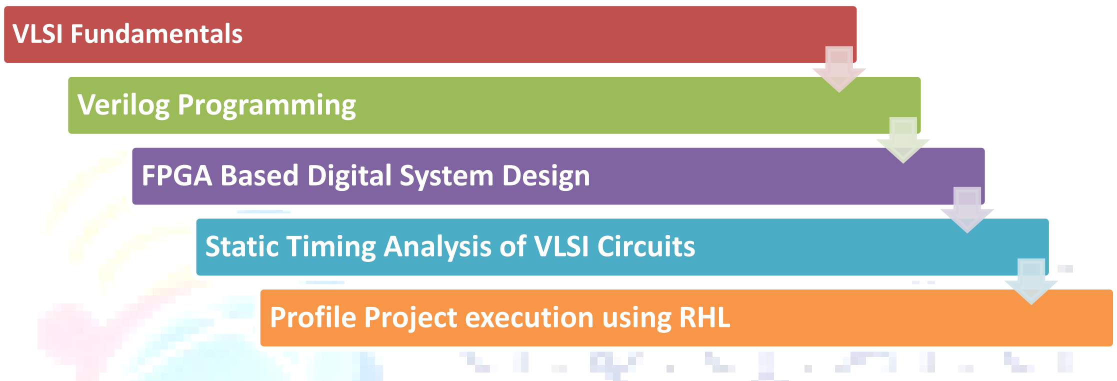
The objective of O-Level chip design program is to introduce the participants to the necessary skills and knowledge to excel in the field of Physical Design and FPGA emulation. The program is of immediate relevance to industry and makes the participants exactly suitable for the VLSI Industry

Outcome of the Course:

After successful completion of this Course, students will be able to:

- Design and develop IPs for VLSI using Verilog HDL and prototype them on FPGAs
- Create their own IP and SoC or FPGA implementation
- Ability to perform static timing analysis, develop timing constraints, and optimize
- designs for timing closure.
- Develop their software application with Zynq APSoC

Full Flow of Course:



Course Structure:

This course contains totally Nine modules. After completing the first Eight modules, the students have to do a six weeks' project using any of the topics studied to earn the PG Program in Embedded and SoC Design certificate.

Code	NOS Name	Duration (in Hours)
VL151-1	VLSI Fundamentals	60
VL151-2	Verilog RTL coding for Synthesis	60
VL151-3	FPGA Architecture and Programming	60
VL151-4	Static Timing Analysis of VLSI Circuits	60
VL151-5	Employability Skills	60
VL151-6	Project Work/OJT	150
Total Duration		**450**

Course Fees

The course fee is **Rs. 24,500/-** including GST. Can be paid as a single instalment or in 2 instalments as given below,

Fee Component	SC-ST Candidate	General/OBC/EWS Candidate	Last Date
Registration Fee	Nil	Rs. 1000/- for others (Adjustable with total fee)	22 Sep 2025
Tuition Fee (Including NSQF Registration & Examination Fee)			
1st Instalment	Nil	11,750/-	24 Sep 2025
2nd Instalment	Nil	11,750/-	24 Nov 2025
Total	Nil	24,500/-	

Registration Fee.

(**Non-Refundable if candidate is selected for admission but did not join and if a candidate has applied but not eligible.)**

SC/ST: Nil
Others: Rs. 1,000/- (Adjustable with Total fee for candidates)

However, the above registration fee shall be refunded in a few special cases as given below

- Candidates are eligible but not selected for admission.
- The course has been postponed, and the new date is not convenient for the student.
- Course cancelled.

Eligibility

- Completed 12th or equivalent in Science with Physics and Maths

(or)

- Completed 2nd Year of 3-Year Diploma in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches after class 10^{th}

Number of Seats: 30-Total *[tentative]*

Category	No. of Seats
SC (15%)	4
ST (7.5%)	2
GENERAL	24
Total	**30**

Note: Seats are allocated based on the merit of the Qualification.

How to Apply?

Candidates can apply online in our website https://nva.nielit.gov.in/. Payment towards the non-refundable registration fee can be paid through any of the following modes:

- Online transaction:
    - **Account Name**: NIELIT CHENNAI
    - **Account No:** 31185720641
    - **Bank name:** State Bank of India (SBI)
    - **Branch:** Kottur (Chennai)
    - **IFSC Code:** SBIN0001669.
- Pay through UPI Mobile Apps

Note: *The Institute will not be responsible for any mistakes done by either the bank concerned or by the depositor while remitting the amount into our account*

Last date of Registration: **22nd Sep 2025**

Registration Procedure:

All interested candidates are required to fill the Registration form online with registration fees before **22nd Sep 2025** with all the necessary information.

Selection Criteria of candidates:

The selection to the course shall be based on the following criteria:

Selection of candidates will be based on their marks in the qualifying examination subject to eligibility and availability of seats.

- The Provisionally Selected Candidates will be informed regarding **selection via email**. In case of vacancy, additional selection lists will be prepared and will be intimated by email only.
- Provisionally selected candidate has to upload their document on registration portal for online verification.

    - ✓ Original Copies of Proof of Age, Qualifying Degree (Consolidated Mark sheet & Degree Certificate/Course Completion Certificate), 10th and 12th mark sheet.
    - ✓ One passport size photograph.
    - ✓ Self-attested copy of Govt. issued photo ID card.

- After document verification, selected candidates have to pay first instalment of **Rs. 11,750/-** or as applicable on or before **24th September 2025** by payment mode mentioned above. Selected candidates are requested to upload the proof of remittance of fee on registration portal and also send the proof of remittance of fee as email to ishant@nielit.gov.in or trng.chennai@nielit.gov.in

Admission:

All provisionally selected candidates whose documents are verified and have paid the fees (full or first instalment) and verified by the accounts section of NIELIT Chennai will get a welcome message for the provisional admission in his/her login ID provided during registration. The credentials and URL for the online portal will be shared through WhatsApp or email.

Note: Provisionally Selected Candidates have to visit NIELIT Chennai for Certificate Verification & Final Examination. Otherwise, their candidature will be cancelled without any intimation.

Discontinuing the course:

- No fees (including the security deposit) under any circumstances, shall be refunded in the event of a student who have completed the process of admission or discontinuing the course in between. No certificate shall be issued for the classes attended. Only Grade Sheet will be issued.
- If candidates are not uploading consecutive 3 assignments within assigned time their candidature will be cancelled without any notice and all fees paid will be forfeited.
- If candidates are not appearing for any internal examinations/practical their candidature will be cancelled without any notice and all fees paid will be forfeited

Course Timings:

This program is a practical oriented one and hence there shall be more lab than theory classes. The cloud based online theory classes will be on forenoon and lab session will be conducted mostly on afternoon time.

Address:

National institute of Electronics and Information Technology
ISTE Complex, No. 25, Gandhi Mandapam Road, Chennai – 600025
E-mail: ishant@nielit.gov.in
Phone: 044-24421445
Contact Person: Mr. Ishant Kumar Bajpai
Mobile: 9445240125 [Call @ 9 AM to 5 PM]

Course enquiries:

Students can enquire about the various courses either on telephone or by personal contact between 9.15 A.M. to 5.15 P.M. (Lunch time 1.00 pm to 1.30 pm) Monday to Friday.

Placement:

Students who have completed the course successfully and qualified, will be given placement guidance and career counselling to crack the interviews.

Important Dates:

- **Last Date of Registration:** 22-09-2025
- **Payment of first instalment fee**: on or before 24-09-2025
- **Course Start Date**: 24-09-2025
- **Payment of second instalment fee:** 24-11-2025

Examination & Certification:

Final Certificates will be issued after successful completion of all the modules including mini project. For getting certificate a candidate has to pass each module individually with minimum required marks of 50%.

Examination Scheme:

Examination scheme for each module is as follows:

Module Name	Total Marks	Written	Practical / Assignment
VLSI Fundamentals	100	25	75
Verilog RTL coding for Synthesis	100	25	75
FPGA Architecture and Programming	100	20	80
Static Timing Analysis of VLSI Circuits	100	25	75
Employability Skills	100	25	75
Project Work/OJT	100	NA	100
Total	600	120	480

Grading Scheme:

- Following Grading Scheme (on the basis of total marks) will be followed:

Grade	S	A	B	C	D	Fail
Marks Range (in %)	85 to 100	75 to 84	65 to 74	55 to 64	50 to 54	Below 50

- Final Grading as per above grading scheme will be given on the basis of total marks obtained in all modules. For last module (ES 609) grade will be given on the basis of project demonstration.

Lab Infrastructure Details:

Hardware Facilities:

- Development Boards - STM32, ARM Cortex-M4
- Sensors–PIR, Ultrasonic, Soil Moisture, Accelerometer & Gyro meter
- FPGA- Xilinx Kintex-KC705 Evaluation Kit, Xilinx Virtex - VC 707 Evaluation Kit, ZedBoard-Zyng-7000 Development Board
- DSP Evaluation Board – TI-EVK2H with Arm Cortex-A15 processor

Software Facilities:

- Xilinx Vivado Design Tool
- Cadence IC Design & Verification Flow
- Synopsis IC Design & Verification Flow

Faculty Profile:

Ishant Kumar Bajpai (Scientist D), extensive experience spans over a decade and revolves around implementing funded projects in two key areas: the Internet of Things (IoT) and Very Large Scale Integration (VLSI) with applications in the Biomedical and Automotive industries. His research interests are focused on FPGA implementations of Digital Signal Processing (DSP) systems.

.

Naresh Raja T (Resource Person – VLSI), expertise in VLSI design, embedded systems, and functional verification. Hands-on experience with Synopsys and Cadence tools, specializing in synthesis, UVM-based functional verification, and physical design. Skilled in embedded system development and SoC architecture, ensuring high-performance, reliable solutions.

Detailed Syllabus of the Course

VL151-1: VLSI Fundamentals

Duration: 60 Hours

Objective:

The objective of the module is to provide a detailed review of VLSI fundamentals for a thorough understanding of the concepts and techniques for analog and digital system design.

Module Description:

Introduction to VLSI Design:

- Overview of VLSI technology and its significance
- Introduction to VLSI design flow: Front-end and Back-end

CMOS Transistor Theory:

- Fundamentals of MOSFET operation
- MOSFET characteristics and modelling.
- MOSFET scaling trends and technology advancements

CMOS Inverter Characteristics:

- CMOS inverter operation and voltage transfer characteristics
- Noise margin analysis and power consumption considerations
- Inverter sizing and optimization techniques

CMOS Logic Design:

- Introduction to basic logic gates: AND, OR, NAND, NOR
- Combinational logic design using CMOS technology
- Designing complex logic functions using CMOS logic gates

Transistor Level Schematics and Layouts

- Transistor level design techniques
- Schematic design and layout considerations
- Design rules, metal layers, and interconnect routing

On-Chip Wire Modeling:

- Introduction to on-chip interconnects
- Wire parasitics and their impact on circuit performance
- Modeling and simulation of on-chip wires

Bonding Diagram, Packaging, and Assembly:

- Overview of packaging technologies and assembly processes
- Bonding diagram design and considerations
- Packaging trends and challenges

Gate Delays and Logical Effort:

- Gate delay modeling and estimation
- Introduction to logical effort analysis
- Determining the best delay-power trade-off for logic gates using P/N ratio

Combinational Logic Circuit Critical Path Optimization:

- Identifying critical paths in combinational circuits
- Gate sizing and logic restructuring techniques
- Timing optimization for improved performance

Timing in Sequential Circuits:

- Introduction to sequential circuits: flip-flops, registers
- Setup and hold time analysis
- Sequential circuit timing considerations and optimization techniques

Learning Outcomes:

The Qualifier of the NOS should be able to:

- Construct digital logic gates using CMOS technology
- Analyze and optimize the performance of digital gates
- Anticipate the impact of parasitics and technology scaling
- Implement a semi-custom integrated circuit from a given RTL code to layout
- Link simplified abstract models to detailed computer simulations

Reading List:

- Design of Analog CMOS ICs - Razavi. The best book available on CMOS analog.
- Microelectronic circuits : Adel Sedra and Kenneth C. Smith
- Franco S, Design with Operational Amplifiers and Analog Integrated Circutis
- CMOS Analog circuit design - Allan and Holberg
- Analog Integrated Circuit Design - Ken Martin and David Johns
- Digital Design by Morris Mano & Michael D Ciletti
- Digital Design: Principles and Practices by John F. Wakerly
- Digital Design by Frank Vahid

VL151-2: Verilog RTL coding for Synthesis

Duration: 60 Hours

Objective:

The objective of the course is to provide a thorough understanding of and hands-onwith digital design & Verification using Verilog HDL

Module Description:

RTL Design Methodology:

- Basics of Register Transfer Level (RTL) design
- Overview of RTL design process and methodology

- Introduction to Hardware Description Languages (HDLs) - Verilog

RTL Design Using HDL:

- Introduction to Verilog syntax and construct
- Designing combinational and sequential logic using HDL
- Writing RTL code for basic digital circuits

RTL Simulation and Verification:

- Functional verification techniques for RTL designs
- Introduction to test benches and test bench development
- Simulation and debugging of RTL designs

Timing Constraints and Analysis:

- Introduction to timing constraints in RTL design
- Timing analysis and optimization techniques
- Setup and hold time violations and resolution

Learning Outcomes:

After successful completion of the module, the students shall be able to:

- Understand the Design Cycle of VLSI.
- Understand Verilog programming syntax, Level of Abstraction in Verilog programming and test bench simulation.
- Design and Develop IPs for VLSI using Verilog
- Emulate, debug & characterize reusable IPs

Reading List:

- Verilog HDL - A guide to Digital Design and Synthesis by Samir Palnitkar.
- A Verilog HDL Primer by J.Bhasker.
- Verilog HDL Synthesis, A Practical Primer by J. Bhasker
- Verilog Digital System Design by Zainalabedin Navabi
- Fundamentals Of Digital Logic With Verilog Design by Stephen Brown
- Modeling, Synthesis, and Rapid Prototyping with the VERILOG HDL by Michael D. Ciletti
- The Verilog Hardware Description Language by Donald E. Thomas & Philip R. Moorby
- IEEE Std 1364-2005 : IEEE Standard Hardware Description Language based on the Verilog Hardware Description Language by the IEEE

VL151-3: FPGA Architecture and Programming

Duration: 60 Hours

Objective:

The objective of the course is to provide a thorough understanding about and hands-on practice with FPGA based digital system design and emulation

Description:

Introduction to FPGAs:

- Overview of Field-Programmable Gate Arrays (FPGAs) and their applications

- Advantages of using FPGAs in digital design

FPGA Architecture and Components:

- Understanding the internal architecture of FPGAs
- Basic components of an FPGA: Look-Up Tables (LUTs), flip-flops, interconnects, etc.
- Introduction to FPGA families and resources

FPGA Design Flow:

- Overview of the FPGA design flow and methodology
- Design entry, synthesis, placement, routing, and bit-stream generation
- Introduction to design constraints

FPGA Design Optimization Techniques:

- Timing constraints and analysis in FPGA designs
- Pipelining, retiming, and other optimization techniques
- Trade-offs between area, power, and performance

Learning Outcomes:

After successful completion of this module, students should be able to:

- Construct digital logic circuits using Verilog HDL
- Simulate those designs using industry-standard simulation tools
- Synthesis and port the design onto FPGA using industry-standard tools
- Implement a project of good complexity onto FPGA
- Debug the FPGA Designs with advanced FPGA tools.

Reading List

- FPGA-Based System Design by Wayne Wolf
- Advanced FPGA Design Architecture, Implementation and optimization by Kilts
- Xilinx® User guides & documentation available at https://www.xilinx.com/support
- /documentation-navigation/overview.html
- Intel® FPGA User guides & documentation available at https://www.intel.com/content/www/us/en/programmable/documentation/lit-index.html
- Swadeshi Microprocessors user guides & documentation available at available at https://shakti.org.in/
- Embedded Core Design with FPGAs. by Zainalabedin Navabi
- FPGA Prototyping by Verilog Examples by Pong P. Chu
- Digital Design Using Digilent FPGA Boards Verilog / Vivado Edition RichardE Haskell, Darrin M Hanna
- FPGA Design: Best Practices for Team-Based Reuse by Philip AndrewSimpson

---

VL151- 4: Static Timing Analysis of VLSI Circuits

Duration: 60 Hours

Objective:

The objective is to provide students and professionals with a comprehensive understanding of how to analyze and ensure the timing correctness of digital VLSI designs without relying on dynamic simulations.

Description:

Overview:

- Introduction to Timing Analysis
- Combinational circuit timing- races and hazards, sequential circuit timing-set up and hold timing, maximum frequency of operation,
- Practical examples of setup and hold time violations and their solutions, timing constraints for synthesis, circuit synthesis, and timing analysis.

Timing performance:

- Techniques to improve timing performance, including pipeline, retiming etc.
- Clock skew, Open loop and closed loop timing. Block level and chip level timing analysis.

STA using EDA tools:

- Static timing analysis using EDA tools.
- Timing analysis after synthesis and place and route, Timing report analysis.
- ECO flow and sign off checks. Timing closure.

Learning Outcomes:

After successful completion of the module, the students shall be able to:

- Explain core timing concepts in digital circuits such as propagation delay, setup and hold time, clock skew, clock jitter, and timing windows.
- Perform static timing analysis to evaluate the timing behavior of combinational and sequential circuits without dynamic simulation.
- Interpret and apply timing constraints using SDC and understand the role of timing libraries (e.g., .lib files).
- Identify and analyze critical and non-critical paths in a digital design and assess their impact on overall performance.

Reading List:

- *"Static Timing Analysis for Nanometer Designs"*
  By: J. Bhasker and Rakesh Chadha A comprehensive and practical book focused entirely on STA with real-world examples and deep insights into timing concepts.
- *"Digital Integrated Circuits: A Design Perspective"* (2nd Edition)
  By: Jan M. Rabaey, Anantha Chandrakasan, and Borivoje Nikolić Covers foundational digital circuit design and includes chapters on timing, clocking, and performance.
- "*CMOS VLSI Design: A Circuits and Systems Perspective*" (4th or 5th Edition)
  By: Neil H. E. Weste and David Harris Offers a broad view of VLSI design with dedicated content on timing, delays, and STA.

VL151-5: Employability Skill

Duration: 60 Hours

Description:

Constitutional Values – Citizenship:

- Explain constitutional values, including civic rights and duties, citizenship, responsibility towards society, and personal values and ethics such as honesty, integrity, caring, and respecting others that are required to become a responsible citizen.
- Demonstrate how to practice different environmentally sustainable practices.
- Becoming a Professional in the 21st Century.
- Discuss relevant 21st-century skills required for employment.
- Highlight the importance of practicing 21st-century skills like Self-Awareness, Behavior Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn, etc., in personal or professional life.
- Create a pathway for adopting a continuous learning mindset for personal and professional development.

Basic English Skills:

- Show how to use basic English sentences for everyday conversation in different contexts, in person and over the telephone.
- Read and understand text written in basic English.
- Write a short note/paragraph/letter/e-mail using correct basic English.
- Career Development & Goal Setting
- Create a career development plan.
- Identify well-defined short- and long-term goals.

Communication Skills:

- Demonstrate how to communicate effectively using verbal and nonverbal communication etiquette.
- Write a brief note/paragraph on a familiar topic.
- Explain the importance of communication etiquette, including active listening for effective communication.
- Role-play a situation on how to work collaboratively with others in a team.
- Diversity and Inclusion
- Demonstrate how to behave, communicate, and conduct appropriately with all genders and PwD.
- Discuss the significance of escalating sexual harassment issues as per the POSH act.

Financial and Legal Literacy:

- Discuss various financial institutions, products, and services.
- Demonstrate how to conduct offline and online financial transactions safely and securely and check passbook/statement.
- Explain the common components of salary such as Basic, PF, Allowances (HRA, TA, DA, etc.), tax deductions.
- Calculate income and expenditure for budgeting.
- Discuss legal rights, laws, and aids.

Essential Digital Skills:

- Describe the role of digital technology in day-to-day life and the workplace.
- Demonstrate how to operate digital devices and use the associated applications and features safely and securely.
- Demonstrate how to connect devices securely to the internet using different means.
- Follow the dos and don’ts of cybersecurity to protect against cybercrimes.
- Discuss the significance of displaying responsible online behavior while using various social media platforms.
- Create an email id and follow email etiquette to exchange emails.
- Show how to create documents, spreadsheets, and presentations using appropriate applications.
- Utilize virtual collaboration tools to work effectively.

Entrepreneurship:

- Explain the types of entrepreneurship and enterprises.
- Discuss how to identify opportunities for potential business, sources of funding, and associated financial and legal risks with its mitigation plan.
- Describe the 4Ps of Marketing-Product, Price, Place, and Promotion and apply
- them as per requirement.
- Create a sample business plan for the selected business opportunity.

Customer Service:

- Classify different types of customers.
- Demonstrate how to identify customer needs and respond to them in a professional manner.
- Discuss various tools used to collect customer feedback.
- Discuss the significance of maintaining hygiene and dressing appropriately.

Getting ready for Apprenticeship & Jobs:

- Draft a professional Curriculum Vitae (CV).
- Use various offline and online job search sources to find and apply for jobs.
- Discuss the significance of maintaining hygiene and dressing appropriately for an interview.
- Role-play a mock interview.
- List the steps for searching and registering for apprenticeship opportunities.

VL151-6: Project Work/OJT

Duration: 150 Hours

Description:

The students can select hardware, software, or system-level projects. The project can be implemented using FPGAs, & IC Design tools, which students have studied and used during the course.

****